

Harsh Patel

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Aspiring Software Engineer from an Electronics Engineering background with a strong interest in full-stack development and backend systems. Experienced in building and deploying production-grade applications using React.js, Node.js, and MongoDB, with exposure to AI integration via Google Gemini. Passionate about writing clean, maintainable code with well-defined architecture — from layered REST APIs to structured frontend patterns.

Education

Kamla Nehru Institute Of Technology, Sultanpur

B.Tech – Electronics Engineering

CGPA: 7.79

Sep. 2022 – May 2026

Sultanpur, Uttar Pradesh

Relevant Coursework: Data Structures and Algorithms | Object-Oriented Programming | Operating Systems

Projects

[Resume Match AI](#) · [Live](#)  | *React.js, Node.js, Express.js, MongoDB, Gemini AI, Zod, SCSS* 2026

- Built and deployed a full-stack AI-powered platform that accepts a resume PDF, job description, and self-description as input, then generates a complete interview preparation package — match score, technical questions, behavioral questions, skill gap analysis, preparation roadmap, and an ATS-optimized resume PDF.
- Integrated Google Gemini 2.5 Flash via the `@google/genai` SDK with Zod schema validation and zod-to-json-schema to enforce strict structured AI output contracts, eliminating hallucinated or malformed responses from the model.
- Implemented secure JWT authentication using `httpOnly` cookies with a MongoDB token blacklist (24-hour TTL index) for logout invalidation; passwords hashed with `bcryptjs`; all report endpoints scoped to authenticated user to prevent cross-user data access.
- Architected the backend with a strict layered structure (controllers, services, models, routes, middleware, templates) and the frontend with a 4-layer pattern (UI → Hook → Context → API Service); built a Puppeteer-based PDF generation pipeline that converts AI-structured JSON into a print-ready A4 resume via an HTML template.

[ThreadForge](#) · [Live](#)  | *React.js, TypeScript, Node.js, Express.js, MongoDB, Tailwind CSS* 2026

- Built and deployed a full-stack discussion platform with a React.js + TypeScript frontend (Vercel) and a Node.js + Express.js REST API backend (Render), backed by MongoDB; features JWT-based authentication, protected routes, and CRUD operations across all resources.
- Designed and implemented a recursive comment system supporting unlimited nested discussions, including tree-based rendering, parent-child relationships, soft deletion, and thread preservation; backend structured with a layered architecture across controllers, services, models, routes, and types.
- Built a custom engagement-driven credit reward engine using arithmetic progression logic, with automatic credit allocation and rollback mechanisms; integrated React.js frontend with backend REST APIs with end-to-end API integration.

[Custom Shell Interpreter \(CodeCrafters\)](#) | *Python, Linux, Git, CLI* 2024 – Present

- Built a Unix-like shell in Python as part of the CodeCrafters challenge, implementing core shell behavior from scratch.
- Implemented built-in commands, external command execution through PATH lookup, command history, and tab-completion functionality.
- Developed support for pipelines, I/O redirection, quote handling, and escape sequence parsing using structured command parsing logic.
- Gained hands-on experience with process management, file descriptors, subprocess handling, and terminal-based application development.

Technical Skills

Languages: Python, JavaScript, TypeScript, HTML5, CSS3, SCSS

Developer Tools: Git, GitHub, Linux, VS Code, Postman, Bruno, npm, Vercel, Render

Technologies/Frameworks: React.js, Node.js, Express.js, MongoDB, REST API, Tailwind CSS, Gemini AI, Zod, JWT, MVC

Coursework: Data Structures and Algorithms, Object-Oriented Programming, Operating Systems

Professional Skills: Problem Solving, Debugging, Communication